

Maximum Likelihood

Advanced Statistics Lab

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Definition

Optimization asked:

$$\hat{\theta} = \arg \min_{\theta} f(\theta)$$

Maximum likelihood chooses $f(\theta)$ from a probability model.

Likelihood And Log-Likelihood

For independent observations:

$$L(\theta; \mathbf{y}) = \prod_{i=1}^n p(y_i \mid \theta)$$

$$\log L(\theta; \mathbf{y}) = \sum_{i=1}^n \log p(y_i \mid \theta)$$

In R

`optim()` minimizes.

So we return the negative log-likelihood:

$$-\sum_{i=1}^n \log p(y_i \mid \theta)$$

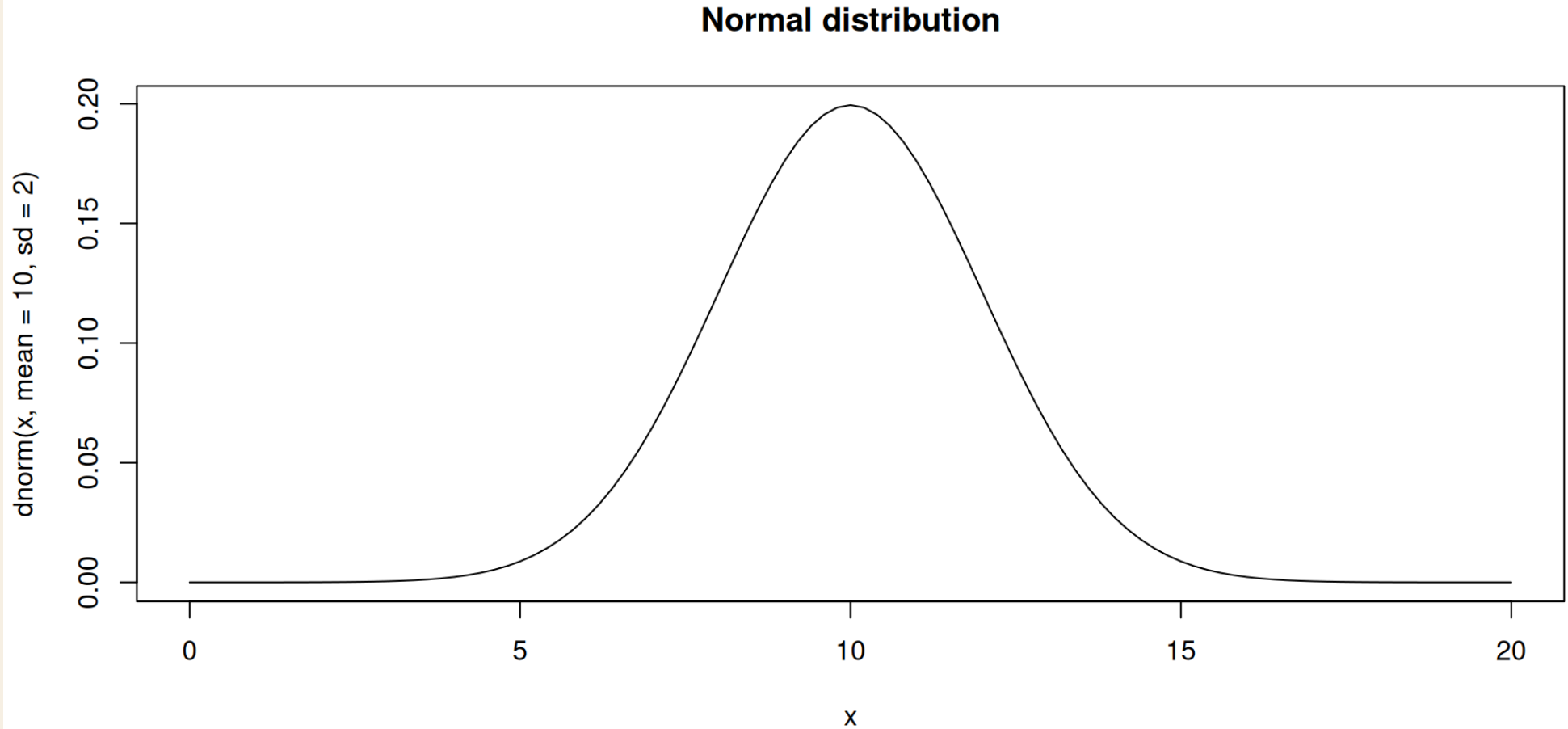
Normal Distribution

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{y - \mu}{\sigma} \right)^2 \right]$$

```
library(ggplot2)
library(knitr)
theme_set(theme_minimal(base_size = 16))

curve(
  dnorm(
    x,
    mean = 10,
    sd = 2
  ),
  from = 0,
  to = 20,
  main = "Normal distribution"
)
```

Normal Density Plot



Finding The Mean

```
set.seed(2026)

mean = 4.5
sd = 1
n = 1000

y = rnorm(n, mean, sd)

lik <- function(par, y, sd = 1) {
  mean = par[1]
  L <- dnorm(x = y, mean = mean, sd = sd, log = TRUE)
  return(-sum(L))
}
```

Estimate Mean

```
p <- c(0)
names(p) <- c("mu")

ML <- optim(p, lik, y = y)

data.frame(
  method = c("Maximum Likelihood", "Empirical"),
  mu = c(ML$par, mean(y))
)
```

	method	mu
mu	Maximum Likelihood	4.514063
	Empirical	4.513767

Finding Mean And Standard Deviation

```
sdlik <- function(x) {  
  res <- 0  
  if (x > 0) res = 1  
  return(log(res))  
}  
  
lik <- function(par, y) {  
  mean = par[1]  
  sd = par[2]  
  
  L2 <- sdlik(sd)  
  if (!is.finite(L2)) return(Inf)  
  
  L1 <- dnorm(x = y, mean = mean, sd = sd, log = TRUE)  
  return(-sum(L1 + L2))  
}
```

Estimate Both

```
p <- c(0, 1)
names(p) <- c("mu", "sigma")

ML <- optim(p, lik, y = y, method = "Nelder-Mead")

rbind(
  ML = ML$par,
  empirical = c(mean(y), sd(y))
)
```

	mu	sigma
ML	4.513704	0.9893471
empirical	4.513767	0.9901296

Estimating Correlation

```
library(mvtnorm)

corlik <- function(x) {
  res <- 0
  if (x >= -1 & x <= 1) res = 1
  return(log(res))
}

varlik <- function(x) {
  res <- 0
  if (x > 0) res = 1
  return(log(res))
}
```

Build Sigma

```
lik <- function(par, y) {  
  L2 <- corlik(par[5])  
  L3 <- varlik(par[3])  
  L4 <- varlik(par[4])  
  if (!all(is.finite(c(L2, L3, L4)))) return(Inf)  
  
  covAB <- par[5] * (sqrt(par[3]) * sqrt(par[4]))  
  sigma <- array(c(par[3], covAB, covAB, par[4]), dim = c(2, 2))  
  
  L1 <- tryCatch(  
    dmvnorm(x = y, mean = c(par[1], par[2]), sigma = sigma, log = TRUE),  
    error = function(e) -Inf  
  )  
  if (!all(is.finite(L1))) return(Inf)  
  
  return(-sum(L1 + L2 + L3 + L4))  
}
```

Estimate Correlation

```
methods = c("Nelder-Mead", "BFGS", "CG", "L-BFGS-B", "SANN", "Brent")

p <- c(0, 0, 1, 1, 0.2)
names(p) <- c("mu1", "mu2", "tauA", "tauB", "rho")

set.seed(2026)
n = 10
tauA <- 3
tauB <- 10
rho = 0.7
covAB <- rho * (sqrt(tauA) * sqrt(tauB))
muA <- 5
muB <- 20
sigma <- array(c(tauA, covAB, covAB, tauB), dim = c(2, 2))
Y <- rmvnorm(n, c(muA, muB), sigma)

ML <- optim(p, lik, y = Y, method = methods[1])

res <- as.list(ML$par)
res$covAB <- res$rho * (sqrt(res$tauA) * sqrt(res$tauB))
```

Correlation Results

```
out <- rbind(  
  unlist(res),  
  c(colMeans(Y), var(Y[, 1]), var(Y[, 2]), cor(Y)[1, 2], cov(Y)[1, 2])  
)  
colnames(out) <- names(res)  
rownames(out) <- c("ML", "empirical")
```

out

	mu1	mu2	tauA	tauB	rho	covAB
ML	-22.484565	7.335499	70.379671	14.028532	0.8906419	27.985487
empirical	4.069481	18.482726	1.771725	9.252332	0.5003696	2.025885

Animal Model Likelihood

$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{a} + \mathbf{e}$$

$$\mathbf{y} \sim MVN(\mathbf{X}\mathbf{b}, \mathbf{A}\sigma_a^2 + \mathbf{I}\sigma_e^2)$$

Animal Likelihood

```
set.seed(2026)
n_animals <- 20
A <- diag(n_animals)
A[1, 2] <- A[2, 1] <- 0.5
A[3, 4] <- A[4, 3] <- 0.5
A[5, 6] <- A[6, 5] <- 0.25

X <- matrix(1, n_animals, 1)
b <- 30
var_a <- 4
var_e <- 9

a <- as.numeric(rmvnorm(1, mean = rep(0, n_animals), sigma = A * var_a))
y <- as.numeric(X %*% b + a + rnorm(n_animals, sd = sqrt(var_e)))
```


Animal Likelihood Function

```
lik <- function(par, y, X, A) {  
  L2 <- varlik(par[1])  
  L3 <- varlik(par[2])  
  if (!all(is.finite(c(L2, L3)))) return(Inf)  
  
  b <- matrix(c(par[3:(2 + ncol(X))]), ncol = 1)  
  G <- A * par[2]  
  R <- diag(length(y)) * par[1]  
  V <- G + R  
  mu <- X %*% b  
  
  L1 <- tryCatch(  
    dmvnorm(x = y, mean = as.numeric(mu), sigma = V, log = TRUE),  
    error = function(e) -Inf  
  )  
  if (!all(is.finite(L1))) return(Inf)  
  
  return(-sum(L1 + L2 + L3))  
}
```

Estimate Animal Model

```
getp <- function(X, b = 0.1, var_e = 1, var_a = 1) {
  p <- c(var_e, var_a, rep(b, ncol(X)))
  names(p) <- c("var_e", "var_a", paste("b", 0:(ncol(X) - 1), sep = ""))
  p
}

ML <- optim(getp(X, b = mean(y)), lik, X = X, A = A, y = y, method = methods[1])

rbind(
  ML = ML$par,
  true = c(var_e, var_a, b)
)
```

	var_e	var_a	b0
ML	8.290905	7.775783e-06	30.09605
true	9.000000	4.000000e+00	30.00000

Take-Home

Maximum likelihood is not a new algorithm.

It is an optimization problem where the objective comes from a probability model.

The code pattern is:

1. collect unknowns in `p`,
2. build the likelihood in `lik()`,
3. run `optim(p, lik, ...)`.

Exercises

1. Change starting values in **p**.
2. Estimate **mu** and **sigma** with $n = 20$.
3. Change the true value of **rho**.
4. Change the relationship matrix **A**.